

Connected Care and Hybrid Quantum-Classical Research

Role-based portals, calibrated comparisons, explainable reports and multilingual assistance

Updated 3 October 2026. Navy and teal design. Workspace: C:/Users/strka/Downloads/mlp.

Evidence source	Scope
Implementation	React/TypeScript + modular FastAPI services
Saved experiment	b147c20b14494a3fbc457714046b8683
Public dataset	UCI Cleveland Heart Disease
Method	5 folds repeated twice; separate final test
Use boundary	Research prototype; public/anonymized/synthetic data

No diagnosis, treatment benefit, clinical validity or quantum advantage is claimed. All numerical results below are read from the local database.

1. System difference and scope

The earlier shared dashboard offered single-holdout experiments. The upgraded system adds authenticated patient, doctor, administrator, ambulance-driver and receiving-hospital portals; ownership, assignments, consent, report review and content-free audit metadata. The navy and teal interface retains the research workflow while presenting participant results in plain language.

Earlier system	Implemented upgrade
Shared local dashboard	Revocable sessions, five roles and scoped data
Single held-out split	Repeated development CV plus separate final testing
Uncalibrated probabilities	Sigmoid/isotonic calibration, Brier score and reliability curve
Single-model reports	Classical/quantum consensus and conformal class sets
Global importance	Bounded calibrated SHAP with explicit global fallback
Research results only	Participant reports, doctor notes, what-if and PDF
English interface	Nine catalogs, core translations and Arabic RTL

Simulation remains classical. Hardware execution, prospective clinical validation and production governance are outside the verified scope. Specialized language copy retains English fallback and requires professional review.

2. Architecture and role permissions

React portal -> same-origin /api -> session/role middleware -> ownership-scoped services -> SQLite records, validated CSV datasets and trusted joblib checkpoints. Training uses a single worker; report explanation uses a bounded separate process.

Role	Allowed workflow
Patient	Own profile, consent, measurements, reports/PDF, what-if and own chat history
Doctor	Assigned participants/reports; notes, approvals and explicit overrides; owned experiments and consensus lab
Administrator	Approve staff, assign patients, verify referrals and dispatch fleet; audit/research
Ambulance driver	Assigned transport only; accept, pickup, destination, location and handover
Receiving hospital	Selected referrals; explicit acceptance, preparation, capacity and admission

Argon2 hashes protect passwords. HttpOnly SameSite access cookies last 15 minutes; refresh cookies rotate and expire after seven days. Persisted sessions permit logout revocation. JavaScript does not read credentials.

Modules: emergency.py, photos.py, auth.py, clinical.py, evaluation.py, quantum.py, consensus.py, reports.py, explain.py, whatif.py, chatbot.py, pdf_export.py, repository.py and i18n/. The legacy campus source remains preserved.

3. Datasets and evaluation design

Public benchmark	Rows	Features	Caution
UCI Parkinsons Voice	195	22	Repeated subject recordings: grouped validation required
UCI Cleveland Heart Disease	303	13	Benchmark evidence only
Wisconsin Breast Cancer	569	30	Benchmark evidence only

Uploads are bounded binary-label CSVs. Target mapping is stored explicitly. Exact duplicate rows are removed before splitting. Imputation, standardization, categorical encoding, feature reduction and angle scaling fit inside proper training partitions.

The complete demo uses Cleveland Heart, seed 42, 25% final holdout, two PCA components, equal row/feature budgets, sigmoid calibration, 5 folds repeated twice and three VQC seeds with 120 optimizer evaluations each.

Development is separated from conformal calibration. Within each fold, separate training-side rows fit probability calibration. Model-pair selection uses development CV AUC. The final test is reserved for evaluation.

CV intervals are descriptive t intervals over correlated folds, not independent-sample guarantees. Conformal outputs are label sets, not probability confidence intervals. Independent and grouped validation remains necessary.

4. Measured repeated-CV results

Ten development folds. Each cell shows mean $\pm$ standard deviation and descriptive 95% interval. Values are proportions.

Model	Accuracy	Sensitivity	Specificity	ROC-AUC
Logistic regression	0.798 $\pm$ 0.060 [0.755, 0.841]	0.733 $\pm$ 0.135 [0.636, 0.830]	0.851 $\pm$ 0.094 [0.783, 0.918]	0.895 $\pm$ 0.047 [0.861, 0.928]
Support vector machine	0.820 $\pm$ 0.059 [0.779, 0.862]	0.782 $\pm$ 0.114 [0.701, 0.863]	0.851 $\pm$ 0.063 [0.806, 0.897]	0.879 $\pm$ 0.040 [0.851, 0.908]
Random forest	0.787 $\pm$ 0.054 [0.748, 0.826]	0.745 $\pm$ 0.140 [0.645, 0.846]	0.821 $\pm$ 0.080 [0.763, 0.878]	0.862 $\pm$ 0.057 [0.822, 0.903]
Gradient boosting	0.754 $\pm$ 0.063 [0.709, 0.799]	0.684 $\pm$ 0.138 [0.585, 0.783]	0.810 $\pm$ 0.122 [0.722, 0.897]	0.876 $\pm$ 0.055 [0.836, 0.915]
Quantum kernel	0.627 $\pm$ 0.066 [0.580, 0.675]	0.504 $\pm$ 0.210 [0.354, 0.654]	0.729 $\pm$ 0.146 [0.624, 0.833]	0.701 $\pm$ 0.071 [0.650, 0.752]
Variational quantum classifier	0.787 $\pm$ 0.065 [0.741, 0.834]	0.733 $\pm$ 0.148 [0.627, 0.839]	0.831 $\pm$ 0.082 [0.772, 0.890]	0.853 $\pm$ 0.069 [0.804, 0.903]



CV mean ROC-AUC; whiskers: descriptive 95% interval
The quantum kernel does not outperform classical models in this run. These observed scores are not evidence of clinical validity or a general quantum advantage.

5. Final test, consensus and seed spread

Model	Accuracy	ROC-AUC	Brier	Base / test rows
Logistic regression	0.8553	0.9631	0.0996	135 / 76
Support vector machine	0.8553	0.9436	0.1107	135 / 76
Random forest	0.8421	0.9202	0.1371	135 / 76
Gradient boosting	0.8289	0.9063	0.1255	135 / 76
Quantum kernel	0.6447	0.7366	0.2202	135 / 76
Variational quantum classifier	0.8947	0.9310	0.1215	135 / 76

Matched-holdout diagnostic	Measured value
Agreement rate	0.6842105263157895
Disagreement cases	24
Error rate when agreeing	0.1346153846153846
Error rate when disagreeing	1.0
Disagreement/error correlation	0.8185201195476004

These are exploratory associations. Review routing is not validated clinical triage. Ambiguous or empty conformal sets prompt research review rather than invented probability certainty.

Measured VQC member training losses: 0.44213, 0.44311, 0.41624. Loss standard deviation: 0.01244. These are optimization losses, not performance confidence intervals.

6. Quantum and explainable reports

The input-dependent feature map applies RY rotations, neighboring ZZ phases based on feature products and repeated encoding. Squared state overlaps form an SVM kernel. A unit test confirms that it differs from the product-of-cosine-squared kernel. Exact simulation remains a classical calculation.

VQC uses trainable RY rotations and ring CNOT gates, binary cross-entropy and seeded COBYLA optimization. Independent members are averaged, with measured member losses retained. Circuits are bounded to 2-6 qubits and 1-4 configured layers.

Consent-gated measurements must match the chosen schema and contain finite numerical values. Observed-data range warnings are informational. The report preserves both calibrated outputs, model versions and consensus facts. Doctor notes, approvals and overrides append review events without changing model probabilities.

Kernel SHAP explains the actual calibrated pipeline with a small background and an eight-second worker budget. Tree SHAP is a separate uncalibrated-component diagnostic where supported. Timeouts produce an explicitly labeled global permutation fallback.

Patient summaries use localized simple safety templates and top-feature descriptions. Optional AI narrative is accepted only after number checks. What-if sliders recompute both models without rewriting the report. Counterfactuals are bounded one-feature grid candidates, not guaranteed global minima or medical advice.

7. Languages, assistant and privacy

Nine locale catalogs support English, Tamil, Hindi, Telugu, Malayalam, Kannada, Spanish, French and Arabic. Core portal labels and patient safety templates are translated. Specialized research descriptions retain English fallback. Arabic switches direction and layout. Professional language review remains necessary.

The server-side Anthropic adapter requires an environment API key, model name and explicit external-AI enable flag. The default returns labeled local guidance. Only authorized report facts and fixed platform guidance enter provider context; identities and review notes are omitted.

Language routing, emergency/medication/injection phrase guards, input bounds and rate limits apply before generation. Provider output is buffered and checked before SSE chunks are shown. These guardrails are prototype controls, not a medical safety guarantee.

Chat history is owner-scoped and deletable. Audit entries retain action/resource metadata rather than conversation content. Browser voice input/output is optional and permission-dependent; voice recognition and installed-language voices remain unverified.

Audience PDF exports enforce report ownership, use Unicode/shaping support and reject unsupported script fonts rather than silently rendering missing glyphs. Some technical headings remain English.

8. API and security

Endpoint group	Operations
/api/auth/*	Register, login, refresh, logout, GET/PATCH me
/api/patients; /api/consent	Scoped profiles; GET/PUT research consent
/api/care/*	Separate health consent, assigned consultations and doctor outcome
/api/emergency/*	Scoped cases, transitions, resources, capacity, location and live WebSocket
/api/photos/*	Consent-gated upload and assigned doctor review
/api/admin/*	Users, doctor approval, assignments, audit
/api/datasets; /api/preprocess	Owned research datasets, CSV upload and settings
/api/models/*; /api/experiments/*	Training, metrics, prediction, importance and run status
/api/reports/*	Create, scoped list/detail, review and audience PDF
/api/consensus/lab/{dataset_id}	Measured holdout agreement/error analysis
/api/what-if; /api/what-if/counterfactual	Bounded model exploration
/api/chat; /api/chat/history	Authenticated responses; own GET/DELETE history

Controls include an explicit CORS allow-list, allowed-Origin checks for cookie mutations, HttpOnly cookies, Argon2, rate limits, security headers, bounded inputs and trusted server-generated artifact paths. Secrets and runtime data are excluded from Git and Docker build context.

Use HTTPS and secure cookies outside localhost. SQLite audit triggers are append-only application controls, not tamperproof protection against filesystem administrators. Encryption, retention governance, backups and independent security review remain deployment work.

9. Setup, demo and verification

Requirements: Python 3.10 and Node 22.12+. Run `npm ci` and `python -m pip install -r backend/requirements.lock`. Copy `.env.example` to `.env` and supply a random JWT secret plus a unique strong demo password. Seed with `python -m backend.seed_demo --train --reports`, then run `npm run dev`.

Fictional demo emails: `admin@qura.demo`, `doctor1@qura.demo`, `doctor2@qura.demo`, `patient1@qura.demo`, `patient2@qura.demo`, `patient3@qura.demo`, `driver1@qura.demo`, `driver2@qura.demo`, `hospital1@qura.demo`, `hospital2@qura.demo` and `hospital3@qura.demo`. The password comes only from local `QURA_DEMO_PASSWORD`. Synthetic report measurements are generated; they are not copied patient records.

Check	Recorded outcome
Production frontend build	Passed (Vite 7.3.6)
Vitest frontend/regression tests	33 passed across 3 files
Pytest backend tests	38 passed; isolated temporary storage, including emergency chain, capacity/consent/access and legacy identity migration
Playwright browser flows	15 passed; full five-role emergency UI and existing research/photo flows, fresh temporary storage
Measured demo	Six Heart models; 5 folds repeated twice; separate final holdout
Manual browser check	Login/theme, assigned doctor queue/detail, research evaluation and logout checked
npm audit	5 high transitive findings in retained Tailwind 3 toolchain
Unverified	Live Anthropic, Docker, GPS/device permissions, Web Speech, physical mobile browsers, live traffic/hospital systems, external deployment and clinical validity
Connected workflow	Consultation -> doctor referral -> administrator verification/dispatch -> driver pickup -> hospital acceptance/preparation -> live tracking -> handover/admission
Simulated integrations	Synthetic local map/route links, generated traffic history and manually reported hospital capacity; no real dispatch

Backend tests isolate SQLite and artifacts in temporary storage. Playwright uses a fresh random temporary directory and a reduced two-model smoke configuration. Those scores are not the six-model results reported here.

10. Health profiles and doctor consultation

Patients maintain symptoms, timing/severity notes, history, allergies, medications and contact details. Health sharing uses a separate explicit consent switch. Structured assistant intake saves patient-entered facts without diagnosis. A consultation is routed only to an assigned approved doctor; optional chat sharing requires consent.

The doctor reviews shared facts and records a human outcome: follow-up, examination, care instructions or emergency. Only an emergency outcome permits referral. Research predictions and assistant generation are not required for any transport transition. Consent withdrawal hides the shared health profile from staff.

Photo uploads remain a separate human review workflow. Images are not diagnostic model inputs. Doctors can review only currently assigned consenting patients; drivers and hospitals cannot read research reports, private chat or uploaded images.

11. Connected emergency workflow

Stage	Required human action
Referral	Doctor provides pickup coordinates/address, priority, bed/capability needs and notes
Verification and dispatch	Administrator verifies referral and assigns an available linked ambulance
Pickup	Assigned driver explicitly accepts, travels, confirms pickup and patient onboard
Destination	Driver reviews eligible options; receiving hospital explicitly accepts and reserves a bed
Transport and preparation	Hospital records preparation; driver reports transport and arrival; hospital confirms arrival
Closure	Driver records handover; hospital records admission/bed assignment and closes case

Driver refusal and administrator reassignment release the unit before pickup. Hospital refusal or withdrawal releases the reserved bed and requires another destination. Capacity withdrawal flags accepted cases for explicit reconfirmation. Transactions prevent double fleet/bed allocation. Admission consumes the reservation; the ambulance becomes available.

Every transition checks current session, role, ownership/assignment and valid prior state. Append-only timeline records support review. No hospital is treated as accepting solely because it ranks first.

12. Routing, tracking and simulation boundary

Eligibility considers reported availability, general/ICU bed count, requested capability, a recent capacity report and the configured demo service area. Ranking explains approximate distance and travel time; excluded hospitals have reasons.

The map is a local interactive schematic. Distances use a great-circle approximation with a road multiplier. Linear regression uses 168 synthetic hourly traffic observations and a configured traffic factor. Route links are straight lines, not verified road navigation; no live traffic service is connected.

After driver acceptance, manual demo coordinates or explicitly enabled browser GPS can update the case. Scoped WebSocket events and polling fallback update authorized views, recompute approximate ETA and show stale positions. Permission revocation or reassignment ends access. Device GPS permission and physical mobile hardware remain unverified.

Hospital capacity is manually reported, and seeded fleet/hospitals are fictional. No ambulance service or receiving hospital is contacted. Real dispatch, navigation, traffic, capacity feeds, alert delivery and clinical validation require separately configured integrations and operational review.

13. Limitations and recommended next steps

- No prospective clinical validation, treatment benefit or quantum advantage is established. Exact simulation uses classical hardware; no hardware adapter is configured.
- CV folds overlap. Intervals and conformal sets require careful interpretation and distributional assumptions. Parkinsons repeated recordings require subject-grouped validation.
- SHAP can fall back to global importance. Counterfactuals are bounded and not guaranteed minimal.
- Core translations need professional review; specialized UI/PDF copy may remain English.
- Live Anthropic, Docker, browser voice recognition, mobile browsers and external deployment are unverified. Phrase guards cannot guarantee clinical safety.
- The retained Tailwind 3 toolchain has five high-severity transitive npm audit findings. Vite/Vitest were upgraded; a compatible styling-toolchain migration remains necessary.
- SQLite and append-only triggers are local prototype controls. Encryption, retention policy, production monitoring and an independent security review remain necessary.

Next steps: review remaining translations; migrate the legacy styling toolchain; run grouped and independent external validation; test an explicitly configured provider; and assess governance before shared deployment.

14. References and reproducibility

UCI Cleveland Heart Disease: <https://archive.ics.uci.edu/dataset/45/heart+disease>

UCI Parkinsons Voice: <https://archive.ics.uci.edu/dataset/174/parkinsons>

Wisconsin: scikit-learn load_breast_cancer and its bundled DESCR metadata.

Implementation evidence: README.md, docs/FEATURES.md, docs/VERIFICATION.md, backend modules and src/PortalApp.tsx / src/research/.

Results: backend/storage/research.db; experiment b147c20b14494a3fbc457714046b8683. Joblib checkpoints contain the trusted fitted pipelines.

Rebuild: python tools/build_upgrade_report.py after a completed six-model repeated-CV run.